WORKED SOLUTION

STEP III 2014, Question 8

Pure • Convergence of Series

STEP III 2014, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q8

This question has a full guided treatment in *Beyond STEP*, Section 2.3, *A Divergent Series and Its Convergent Cousin*: https://beyondstep.keithmcnulty.org/ch02/kempner.html

The key idea

Chop the positive integers into blocks $[k^n, k^{n+1})$. Every term in a block lies between the value at the start of the block and the value at the start of the next one, so each block sum is squeezed between the number of terms times those two values. Adding up the blocks turns an awkward series into a geometric series, or into one with a constant term in each block, and we can see at once whether it diverges. The same counting works for the integers with no digit 2, as long as we count how many of them lie in each block.

Solution

The block inequality

The block $k^n \le r \le k^{n+1} - 1$ contains

$$(k^{n+1} - 1) - k^n + 1 = k^{n+1} - k^n = k^n(k-1)$$

integers. Since f is strictly decreasing, for every r in the block,

$$\mathrm{f}(k^{n+1}) < \mathrm{f}(r) \le \mathrm{f}(k^n),$$

the first because $r < k^{n+1}$ and the second because $r \ge k^n$ (with equality only at $r = k^n$). Adding these over the $k^n(k-1)$ values of r gives

$$k^n(k-1)\,\mathrm{f}(k^{n+1}) \le \sum_{r=k^n}^{k^{n+1}-1} \mathrm{f}(r) \le k^n(k-1)\,\mathrm{f}(k^n),$$

as required. (The left-hand inequality is in fact strict.)



The blocks for $k = 2$, with the number of terms in each.

Part (i)

Take $k = 2$ and $f(r) = \frac{1}{r}$, which is strictly decreasing. The block inequality becomes

$$\frac{2^n}{2^{n+1}} \leq \sum_{r=2^n}^{2^{n+1}-1} \frac{1}{r} \leq \frac{2^n}{2^n}, \qquad \text{that is} \qquad \tfrac{1}{2} \leq \sum_{r=2^n}^{2^{n+1}-1} \frac{1}{r} \leq 1.$$

The blocks for $n = 0, 1, \ldots, N$ fit together exactly to cover $1 \leq r \leq 2^{N+1} - 1$, since each block ends just before the next one starts. Adding the $N + 1$ inequalities,

$$\frac{N+1}{2} \leq \sum_{r=1}^{2^{N+1}-1} \frac{1}{r} \leq N + 1.$$

For the deduction: the partial sums $H_m = \sum_{r=1}^{m} \frac{1}{r}$ increase with m, and $H_{2^{N+1}-1} \geq \frac{N+1}{2}$, which can be made larger than any number we like by choosing N large. So the partial sums are unbounded and $\sum \frac{1}{r}$ does not converge.

Part (ii)

Take $k = 2$ and $f(r) = \frac{1}{r^3}$, again strictly decreasing. Now we need only the right-hand inequality:

$$\sum_{r=2^n}^{2^{n+1}-1} \frac{1}{r^3} \leq \frac{2^n}{2^{3n}} = \frac{1}{4^n}.$$

Adding over $n = 0, 1, \ldots, N$ as before,

$$\sum_{r=1}^{2^{N+1}-1} \frac{1}{r^3} \leq \sum_{n=0}^{N} \frac{1}{4^n} < \sum_{n=0}^{\infty} \frac{1}{4^n} = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3}.$$

Ask yourself This bounds only some of the partial sums. Why does it bound the infinite sum?

All the terms are positive, so the partial sums increase. Any partial sum $\sum_{r=1}^{m} r^{-3}$ is at most the one with $2^{N+1} - 1 \geq m$, and so is less than $\frac{4}{3}$. An increasing sequence that is bounded above converges, and its limit is at most the bound. Hence

$$\sum_{r=1}^{\infty} \frac{1}{r^3} \leq \frac{4}{3} = 1\tfrac{1}{3}.$$

Part (iii)

Counting $S(1000)$. Write each integer from 0 to 999 as a string of three digits, padding with leading zeros (so 7 becomes 007). Padding zeros never introduce a 2, so an integer has no 2 exactly when its string has no 2. Each of the three places can be any of the nine digits $0, 1, 3, 4, \ldots, 9$, which gives 9^3 strings, all for different integers. One of them is 000, which is not positive, so $S(1000)$ contains $9^3 - 1$ numbers.

Counting by blocks. The same argument with $n+1$ places shows that there are $9^{n+1} - 1$ integers in $S(10^{n+1})$. So the block $10^n \le r \le 10^{n+1} - 1$ of $(n+1)$-digit numbers contains

$$(9^{n+1} - 1) - (9^n - 1) = 8 \cdot 9^n$$

integers with no 2. (Equivalently, the leading digit has 8 choices, as it cannot be 0 or 2, and each of the other n digits has 9.)

Bounding $\sigma(n)$. This is the block argument with $k = 10$, except that we count only the integers with no 2. Every r in the block has $\frac{1}{r} \le \frac{1}{10^n}$, so the contribution of the block to the sum is at most

$$8 \cdot 9^n \cdot \frac{1}{10^n} = 8\left(\tfrac{9}{10}\right)^n.$$

Now fix n and choose M with $n \le 10^{M+1}$. Every element of $S(n)$ lies in one of the blocks for $m = 0, 1, \ldots, M$, and the reciprocals are all positive. So

$$\sigma(n) \le \sum_{m=0}^{M} 8\left(\tfrac{9}{10}\right)^m < \sum_{m=0}^{\infty} 8\left(\tfrac{9}{10}\right)^m = \frac{8}{1 - \frac{9}{10}} = 80.$$

The second inequality is strict because we left out the positive terms with $m > M$. Hence $\sigma(n) < 80$ for all n.

Checks

- Part (ii): $\sum r^{-3} = 1.2020\ldots$ (Apéry's constant), comfortably below $1\frac{1}{3}$.
- Part (i) with $N = 2$: $\sum_{r=1}^{7} \frac{1}{r} \approx 2.593$, which lies between 1.5 and 3.
- Part (iii): a direct count confirms $728 = 9^3 - 1$ numbers below 1000 and $8 \cdot 9^n$ in each block. Summing directly by computer gives $\sigma(10^7) \approx 10.28$, far below 80. The bound is generous, but it is all we were asked for.

Common slips

- Miscounting the block. It runs from k^n to $k^{n+1} - 1$, which is $k^{n+1} - k^n$ terms, not $k^{n+1} - k^n - 1$.
- Counting all 9^n no-2 strings in the n-th block for (iii). The leading digit cannot be 0, so there are only $8 \cdot 9^n$. Using 9^{n+1} gives a bound of 90, which is not enough.

- Stopping at a bound for the partial sums up to $2^{N+1} - 1$ in (ii) without saying why the whole infinite sum is bounded.

Beyond STEP

This is Cauchy's condensation test: for a decreasing positive sequence, $\sum a_r$ converges exactly when $\sum 2^n a_{2^n}$ does. It shows at once that $\sum r^{-p}$ converges if and only if $p > 1$. Part (iii) is the Kempner series: delete every integer with a given digit and the harmonic series, which diverges, becomes convergent, because almost all large integers contain every digit. Abbott's *Understanding Analysis* covers condensation and the comparison tests carefully.

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